Case Study 3

Immersive Furniture Visualisation for Retail

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1. Application Scenario

A furniture retailer wants a browser-based immersive visualisation tool that lets shoppers inspect a chair and sofa before purchase. The application should present realistic materials, respond convincingly to lighting conditions, and correctly translate user interactions from a logical design space to the screen of a phone or headset.

The demonstration focuses on three graphics problems: selecting an illumination model for a velvet chair, adding believable leather detail to a sofa without increasing mesh complexity, and mapping interaction coordinates to a device viewport.

2. Illumination and Colour Response

Problem: How should a red velvet chair be shaded so that it appears naturally lit in a bright indoor environment?

A Phong-style lighting calculation combines three components. Ambient light provides a baseline illumination so shadowed regions remain visible. Diffuse reflection depends on the angle between the surface normal and the incoming light direction, producing the main shaped appearance of the fabric. Specular reflection adds a view-dependent highlight and is controlled by the viewer direction, reflected light direction, and a shininess parameter.

For velvet, diffuse reflection should be dominant, while specular reflection should remain restrained; otherwise, the fabric may appear like polished plastic. The simplified model used is:

$$I = I_a k_a + I_p[k_d \max(N \cdot L, 0) + k_s \max(R \cdot V, 0)^n]$$

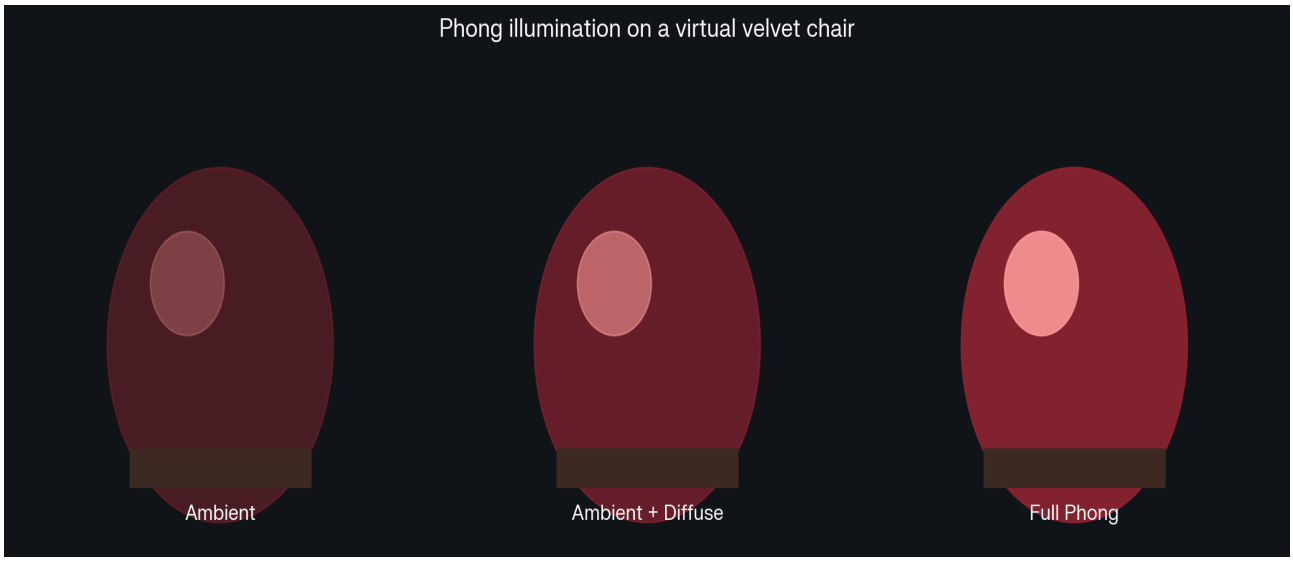


Figure 1. Progressive lighting response: ambient only, ambient plus diffuse, and full Phong shading.

3. Texture and Bump Mapping

Problem: How can a leather sofa look detailed without adding more polygons to its 3D mesh?

Texture mapping wraps a two-dimensional image around the sofa using UV coordinates. The texture can encode leather colour variation, grain, seams, and minor surface imperfections. Bump mapping does not alter the geometry; instead, it perturbs the surface normals used during shading, causing fine grooves and raised details to respond to light.

This approach is efficient for real-time rendering because the mesh vertex count stays low while perceived surface complexity increases. A normal map is a widely used implementation of this idea and generally gives more stable lighting response than a simple height-based bump map.

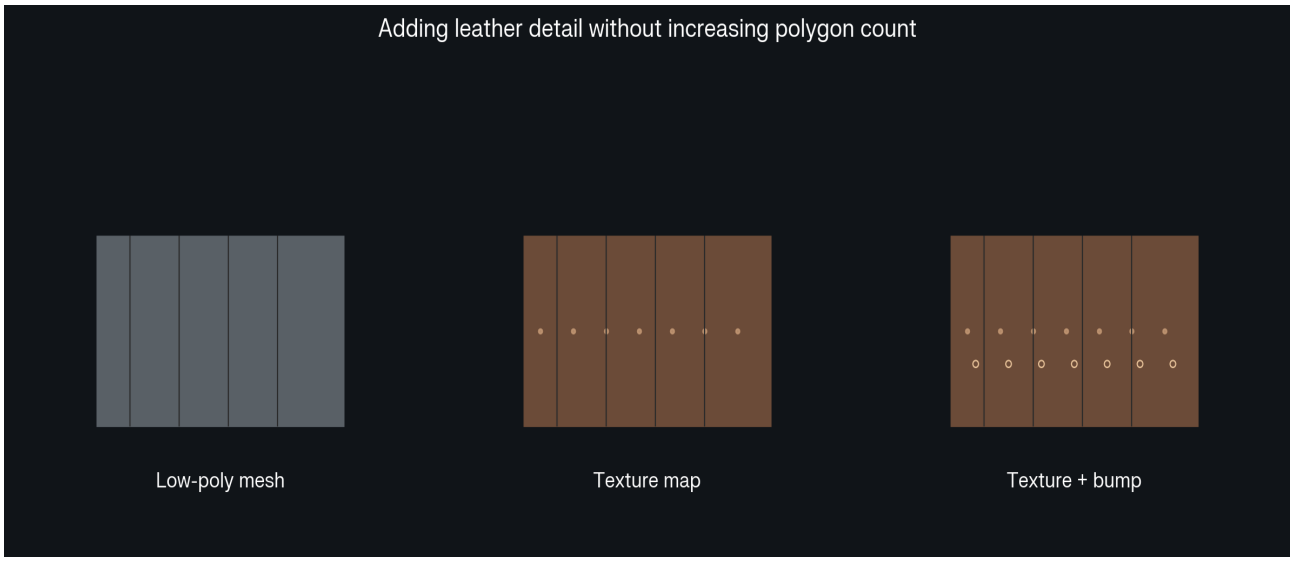


Figure 2. A low-polygon sofa gains apparent leather detail through texture and bump/normal mapping.

4. Window-to-Viewport Transformation

Problem: How can a point selected in a logical interaction space be positioned accurately on a phone display?

After a 3D object is projected, its interaction region can be described in a normalised logical window. That window is then scaled and translated into the pixel bounds of the target viewport. For the x-axis:

$$x_v = x_{v,min} + (x_w - x_{w,min})\,(x_{v,max} - x_{v,min}) / (x_{w,max} - x_{w,min})$$

The y-coordinate uses the equivalent expression. In the demonstration, the logical window is [-1, 1] × [-1, 1] and the phone viewport spans x = 70…1010 and y = 120…1800. Therefore, the logical point (0.25, -0.30) maps to approximately (657.5, 708.0) pixels.

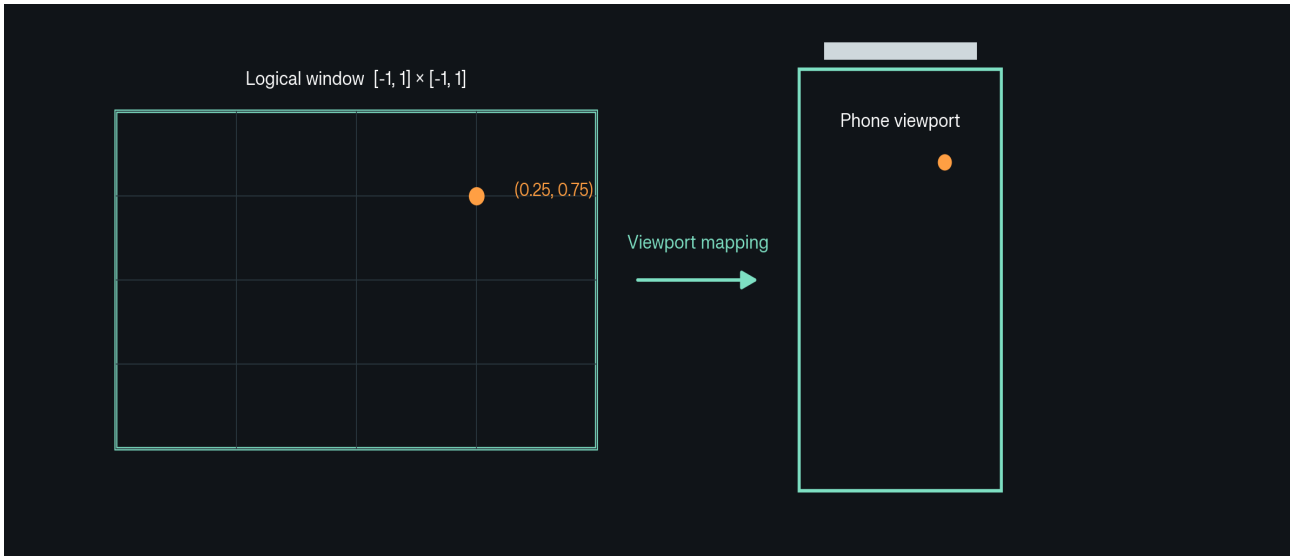


Figure 3. A logical interaction point is mapped proportionally into a smartphone viewport.

5. Rendering Pipeline

The proposed application follows this sequence: camera or headset tracking captures the user's viewpoint; light estimation supplies environmental lighting information; the furniture mesh is loaded with texture and bump/normal maps; Phong shading computes the final surface appearance; projection converts the 3D scene into a 2D image; and viewport mapping aligns interaction controls with device pixels.

Camera tracking → Light estimation → Furniture mesh → Texture and bump maps → Phong shading → Projection → Viewport mapping → Display

6. Performance Considerations

Real-time performance depends on keeping both geometry and shading work manageable. Suitable techniques include low- or medium-polygon furniture models, compressed textures, mipmapping, a limited number of dynamic lights, frustum and occlusion culling, and level-of-detail substitution for distant objects.

Fine visual detail should be moved from geometry into texture and normal maps wherever possible. This reduces vertex processing and memory consumption while preserving a realistic appearance and helping maintain a stable frame rate.

7. User Workflow

A user opens the visualiser, selects a furniture item, and places it in the scene. The user can rotate, scale, and move the object; material controls enable or disable texture and bump mapping; lighting controls adjust ambient, diffuse, and specular response. The rendering pipeline projects the updated scene and maps interaction coordinates to the display.

8. Expected Outcome

The demonstration shows that combining ambient, diffuse, and specular terms produces a more convincing material response than ambient lighting alone. Texture and bump mapping add leather-like detail without increasing polygon count, while window-to-viewport transformation keeps on-screen interaction aligned with the underlying 3D scene.

9. Conclusion

This case study connects core computer-graphics techniques to a practical retail visualisation problem. Illumination modelling improves how virtual furniture responds to light, texture and bump mapping deliver efficient surface realism, and viewport transformation ensures that logical user interactions appear at the correct position on a phone or headset display.

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